

# Introduction to Data Analysis

## What is Data?

Data is a collection of raw facts and figures. It can include numbers, text, images, videos, audio, observations, and measurements. Data alone does not always provide meaning until it is processed and interpreted.

Examples of Data:

- Student marks in an exam
- Sales records of a company
- Temperature measurements
- Social media comments
- Photos and videos

Importance of Data:

- Helps organizations make decisions
- Identifies patterns and trends
- Supports business growth
- Enables prediction and forecasting

## Sources of Data

Data can come from different sources. Mainly there are two types:

Machine-Generated Data:

- Sensor data
- Server logs
- Transaction records
- GPS and IoT devices

Human-Generated Data:

- Social media posts
- Emails and text messages
- Photos and videos
- Survey responses

Modern businesses collect data from both humans and machines to generate useful insights.

## Data Structures

Structured Data:

- Organized in rows and columns
- Stored in databases or spreadsheets
- Easy to analyze

Examples: Excel, SQL databases

Semi-Structured Data:

- Does not follow a fixed schema completely
  - Contains tags or keys
- Examples: JSON, XML, log files

Unstructured Data:

- No fixed format
- Difficult to analyze directly

Examples: Images, audio, emails, videos

Today, organizations work with all three forms of data.

# Types of Data

Nominal Data:

- Categorical data without order

Examples: Gender, Country

Ordinal Data:

- Categorical data with meaningful order

Examples: Education level, Rating

Discrete Data:

- Countable numerical data

Examples: Number of students, Orders

Continuous Data:

- Measurable numerical data

Examples: Height, Weight, Production cost

## What is Information?

Information is processed and organized data that becomes meaningful and useful.

Transformation Process:

Data → Processing → Interpretation → Information

Example:

Raw data: Student marks

Information: Average class performance and pass rate

Good information supports decision-making and business strategy.

# Big Data and its 5Vs

Big Data refers to extremely large and complex datasets that traditional systems cannot process efficiently.

The 5Vs of Big Data:

1. Volume – Large amount of data
2. Variety – Different formats of data
3. Velocity – Speed of data generation
4. Veracity – Quality and reliability of data
5. Value – Business usefulness of data

Examples:

- Facebook data
- E-commerce transactions
- Healthcare records
- Streaming platforms

## What is a Database?

A database is an organized collection of data stored electronically.

Features of Databases:

- Data storage and management
- Fast retrieval of information
- Data integrity and consistency
- Security and authorization
- Scalability

Examples:

- Student management systems
- Banking systems
- E-commerce websites

# Types of Databases

Relational Database (RDBMS):

- Structured tables
- Uses SQL

Examples: MySQL, PostgreSQL

NoSQL Database:

- Flexible structure
  - Handles large-scale data
- Examples: MongoDB, Cassandra

In-Memory Database:

- Stores data in RAM
- Very fast performance

Examples: Redis

Distributed Database:

- Data distributed across multiple systems
- High scalability and availability

# Steps of Data Analysis

Step 1: Understand the Business Problem

Step 2: Collect Data

Step 3: Clean and Prepare Data

Step 4: Analyze Data

Step 5: Visualize Results

Step 6: Communicate Findings

Step 7: Improve and Iterate

These steps help analysts solve real-world business problems systematically.

# Data Analysis vs Data Analytics

Data Analysis:

- Focuses on past and present data
- Finds patterns and trends
- Descriptive and diagnostic approach

Data Analytics:

- Broader field
- Includes prediction and optimization
- Uses machine learning and AI

Both are important for data-driven decision making.

## Types of Analytics

Descriptive Analytics:

- What happened?

Example: Monthly sales reports

Diagnostic Analytics:

- Why did it happen?

Example: Investigating sales decline

Predictive Analytics:

- What may happen?

Example: Predicting customer churn

Prescriptive Analytics:

- What should be done?

Example: Marketing optimization

# Roles in the Data Industry

Data Analyst:

- Cleans and analyzes data
- Creates reports and dashboards

BI Analyst:

- Focuses on business reporting• Uses tools like Power BI and Tableau

Data Scientist:

- Builds predictive models
- Uses AI and machine learning

These roles work together in modern organizations.

# Important Skills for Data Professionals

Technical Skills:

- Python
- SQL
- Excel
- Statistics
- Machine Learning basics

Visualization Tools:

- Power BI
- Tableau
- Looker Studio

Soft Skills:

- Communication
- Problem-solving
- Business understanding

# Career Opportunities

Data professionals are highly demanded worldwide.

Popular Career Paths:

- Data Analyst
- Business Intelligence Analyst
- Data Scientist
- Data Engineer
- Machine Learning Engineer

Benefits:

- High salary
- Remote work opportunities
- Strong career growth
- Global demand

## Conclusion

Data is one of the most valuable assets in the modern world.

Understanding data analysis helps businesses make smarter decisions and solve problems efficiently.

By learning data analysis tools, databases, visualization, and analytics techniques, students can build successful careers in technology and business intelligence.

Thank You!